

Dolžina krivulje

$$\vec{r}(t) = \begin{bmatrix} x(t) \\ y(t) \\ z(t) \end{bmatrix} \quad t \in [a, b]$$

$$\int_a^b \sqrt{\dot{x}^2 + \dot{y}^2 + \dot{z}^2} dt = \int_a^b \|\dot{\vec{r}}(t)\| dt$$

Naravni parameter: $ds = \pm \sqrt{\dot{x}^2 + \dot{y}^2 + \dot{z}^2} dt$

+ ista orientacija
- obratna

Odvod po naravnem parametru:

$$u' = \frac{du}{ds} = \frac{du}{dt} \cdot \frac{dt}{ds} = \frac{\dot{u}}{\dot{s}}$$

(2.) $\vec{r}(t) = \begin{bmatrix} 2t \\ t^2 \\ \frac{4}{3}t^3 \end{bmatrix}$

a) dolžina krivulje od $t=0$ do $t=3$

b) v izhodišču izv. w'' , kjer $w = x^2 + 4y^2 + 9z^2$

a) $l = \int_0^3 \sqrt{4 + 4t^2 + t^4} dt = \int_0^3 (t^2 + 2) dt = \frac{3^3}{3} + 6 = 15$

b) $w = 4t^2 + 4t^4 + t^6 \rightarrow \dot{w} = 8t + 16t^3 + 6t^5 = 2t(3t^4 + 8t^2 + 4) = 2t(t^2 + 2)(3t^2 + 2)$

izhodišče: $t=0$

$$\dot{s} = t^2 + 2$$

$$\rightarrow w' = \frac{\dot{w}}{\dot{s}} = 2t(3t^2 + 2)$$

$$w'' = \frac{\dot{w}'}{\dot{s}} = \frac{6 \cdot 3t^2 + 4}{t^2 + 2} = \frac{18t^2 + 4}{t^2 + 2}$$

$$w'' \Big|_{t=0} = \frac{4}{2} = 2$$

Spremljajoči trieder prostorske krivulje je trojica enotskih vekt.:

- tangenti vektor $\vec{T}$ (smer krivulje, skladna z orientacijo) $\rightarrow$ normala normalne ravnine, smerni vekt. tangente
- normalni vektor $\vec{N}$ (smer spreminjanja $\vec{T}$) $\rightarrow$ normala rektifikacijske ravnine, smerni vekt. normale
- binormalni vektor $\vec{B} = \vec{T} \times \vec{N}$ (normala pritiskljene ravnine)

$$\vec{T} = \frac{\dot{\vec{r}}}{\|\dot{\vec{r}}\|}$$

$$\vec{B} = \frac{\dot{\vec{r}} \times \ddot{\vec{r}}}{\|\dot{\vec{r}} \times \ddot{\vec{r}}\|}$$

$$\vec{N} = \frac{(\dot{\vec{r}} \times \ddot{\vec{r}}) \times \dot{\vec{r}}}{\|(\dot{\vec{r}} \times \ddot{\vec{r}}) \times \dot{\vec{r}}\|}$$

3.

$$\vec{r}(t) = \left(\frac{t^4}{4}, \frac{t^3}{3}, \frac{t^2}{2} \right)$$

a) doloži spr. trieder v t

b) t=1; doloži binormalo in prit. ravn.

$$\dot{\vec{r}} = (t^3, t^2, t)$$

$$\ddot{\vec{r}} = (3t^2, 2t, 1)$$

$$\|\dot{\vec{r}}\| = \sqrt{t^6 + t^4 + t^2} = |t| \sqrt{t^4 + t^2 + 1}$$

$$\dot{\vec{r}} \times \ddot{\vec{r}} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ t^3 & t^2 & t \\ 3t^2 & 2t & 1 \end{vmatrix} = (-t^2, 2t^3, -t^4)$$

$$\vec{T} = \frac{\text{sgn}(t)}{\sqrt{t^4 + t^2 + 1}} (t^2, t, 1)$$

$$\|\dot{\vec{r}} \times \ddot{\vec{r}}\| = t^2 \sqrt{t^4 + 4t^2 + 1} = t^2 \sqrt{t^4 + 4t^2 + 1}$$

$$\vec{B} = \frac{\dot{\vec{r}} \times \ddot{\vec{r}}}{\|\dot{\vec{r}} \times \ddot{\vec{r}}\|} = \frac{1}{\sqrt{t^4 + 4t^2 + 1}} (-1, 2t, -t^2)$$

$$\vec{N} = \vec{B} \times \vec{T} = \frac{\text{sgn}(t)}{\sqrt{(t^4 + t^2 + 1)(t^4 + 4t^2 + 1)}} \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -1 & 2t & -t^2 \\ t^2 & t & 1 \end{vmatrix} = \frac{\text{sgn}(t)}{\sqrt{(t^4 + t^2 + 1)(t^4 + 4t^2 + 1)}} (2t + t^3, 1 - t^4, -t - 2t^3)$$

$$\vec{B}|_{t=1} = \left(-\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, -\frac{1}{\sqrt{6}} \right)$$

$$\vec{r}(1) = \left(\frac{1}{4}, \frac{1}{3}, \frac{1}{2} \right)$$

binormala:

$$\vec{b} = \vec{r}(1) + s \cdot \vec{B}|_{t=1}$$

$$\vec{b} = \begin{bmatrix} \frac{1}{4} - s \\ \frac{1}{3} + 2s \\ \frac{1}{2} - s \end{bmatrix}$$

$$\text{ekv. } s = \frac{1}{4} - x = \frac{1}{2}(y - \frac{1}{3}) = \frac{1}{2} - z$$

prit. ravn.

$$\frac{1}{4} - \frac{2}{3} + \frac{1}{2} = \frac{3 - 8 + 6}{12} = \frac{1}{12}$$

ne rabimo enotske

$$\langle \vec{r} - \vec{r}(1), \vec{B} \rangle = 0$$

$$\left\langle \begin{bmatrix} x - \frac{1}{4} \\ y - \frac{1}{3} \\ z - \frac{1}{2} \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix} \right\rangle = 0$$

$$-(x - \frac{1}{4}) + 2(y - \frac{1}{3}) - (z - \frac{1}{2}) = 0$$

$$-x + 2y - z = -\frac{1}{12}$$

$$x - 2y + z = \frac{1}{12}$$

Ukrivljenosti prost. kriv.

hitrost spreminjanja spr. 3edra merimo z:

- upognjenost κ (spreminjanje $\vec{T}$)
- torzijska ukr./zvitost ω (spreminjanje $\vec{B}$)

Frenet-Serretove formule:

$$\vec{T}' = \kappa \vec{N}$$

$$\vec{N}' = -\kappa \vec{T} + \omega \vec{B}$$

$$\vec{B}' = -\omega \vec{N}$$

$$\kappa = \frac{\|\dot{\vec{r}} \times \ddot{\vec{r}}\|}{\|\dot{\vec{r}}\|^3}$$
$$\omega = \frac{\langle \dot{\vec{r}} \times \ddot{\vec{r}}, \ddot{\vec{r}} \rangle}{\|\dot{\vec{r}} \times \ddot{\vec{r}}\|^2} \quad \leftarrow \text{mešani produkt}$$

④ Parametriziraj krivuljo

$$y = e^z \sin z$$

$$x^2 + y^2 = e^{2z}$$

Pri $z=0$, $x>0$ dobimo κ, ω .

$$x^2 = e^{2z} - e^{2z} \sin^2 z = e^{2z} \cos^2 z$$

$$x = \pm e^z \cos z$$

$\vec{r}(z) = (\pm e^z \cos z, e^z \sin z, z)$ relevantna je + ($x > 0$)

$$\vec{r}(z) = (e^z \cos z, e^z \sin z, z)$$

$$\begin{aligned} \dot{\vec{r}}(z) &= (e^z(\cos z - \sin z), e^z(\sin z + \cos z), e^z) \\ &= e^z(-\sin z, \cos z, 1) \end{aligned}$$

$$\ddot{\vec{r}}(z) = 2e^z(-\sin z - \cos z, \cos z - \sin z, 0)$$

$$\dot{\vec{r}}(0) = (1, 1, 1)$$

$$\dot{\vec{r}} \times \ddot{\vec{r}} = \begin{vmatrix} i & j & k \\ 1 & 1 & 1 \\ 0 & 2 & 0 \end{vmatrix} = (-2, 0, 2)$$

$$\ddot{\vec{r}}(0) = (0, 2, 0)$$

$$\kappa = \frac{\sqrt{4+0+4}}{\sqrt{3^3}} = \frac{2\sqrt{2}}{3\sqrt{3}}$$

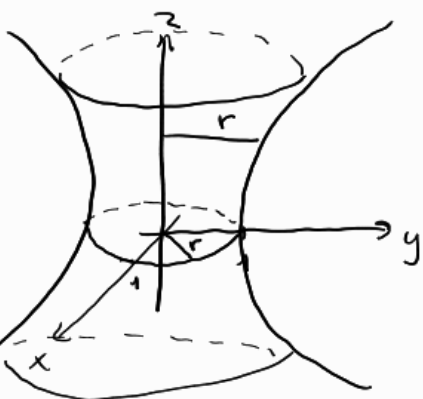
$$\ddot{\vec{r}}(0) = (-2, 2, 0)$$

$$\omega = \frac{\langle (-2, 0, 2), (-2, 2, 0) \rangle}{8} = \frac{4}{8} = \underline{\underline{\frac{1}{2}}}$$

6. Parametriziraj ploskev

$$x^2 + y^2 - z^2 = 1$$

v točki $T(3, 4, z > 0)$ poišči $\vec{N}$, normalo in tang. ravn.



$$x^2 + y^2 = (\sqrt{1+z^2})^2$$

$$\vec{r}(z, \varphi) = \begin{bmatrix} \sqrt{1+z^2} \cos \varphi \\ \sqrt{1+z^2} \sin \varphi \\ z \end{bmatrix} \quad \begin{array}{l} z \in \mathbb{R} \\ \varphi \in [0, 2\pi) \end{array}$$

$$\vec{r}_z = \begin{bmatrix} \frac{z}{\sqrt{1+z^2}} \cos \varphi \\ \frac{z}{\sqrt{1+z^2}} \sin \varphi \\ 1 \end{bmatrix} \quad \vec{r}_\varphi = \begin{bmatrix} -\sqrt{1+z^2} \sin \varphi \\ \sqrt{1+z^2} \cos \varphi \\ 0 \end{bmatrix}$$

$$\vec{r}_z \times \vec{r}_\varphi = \begin{vmatrix} i & j & k \\ \frac{z}{\sqrt{1+z^2}} \cos \varphi & \frac{z}{\sqrt{1+z^2}} \sin \varphi & 1 \\ -\sqrt{1+z^2} \sin \varphi & \sqrt{1+z^2} \cos \varphi & 0 \end{vmatrix} = \begin{bmatrix} -\sqrt{1+z^2} \cos \varphi \\ -\sqrt{1+z^2} \sin \varphi \\ z \end{bmatrix}$$

φ določimo tako, da je:

$$(\cos \varphi, \sin \varphi) = \left(\frac{3}{5}, \frac{4}{5}\right)$$



$$z = -\sqrt{24}$$

$$\vec{N} = \frac{\vec{r}_z \times \vec{r}_\varphi}{\| \vec{r}_z \times \vec{r}_\varphi \|} = \frac{1}{7} \begin{bmatrix} -3 \\ -4 \\ -\sqrt{24} \end{bmatrix}$$

normala: $\vec{R} = \begin{bmatrix} 3 \\ 4 \\ -\sqrt{24} \end{bmatrix} + s \begin{bmatrix} -3 \\ -4 \\ -\sqrt{24} \end{bmatrix} = \begin{bmatrix} 3-3s \\ 4-4s \\ -\sqrt{24}-\sqrt{24}s \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

$$s = \frac{3-x}{3} = \frac{4-y}{4} = \frac{-\sqrt{24}-z}{\sqrt{24}}$$

tang. ravnina

$$\langle \vec{R} - \vec{r}_0, -7\vec{N} \rangle = 0$$

$$\left\langle \begin{bmatrix} x-3 \\ y-4 \\ z+\sqrt{24} \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \\ \sqrt{24} \end{bmatrix} \right\rangle = 0$$

$$3(x-3) + 4(y-4) + \sqrt{24}(z+\sqrt{24}) = 0$$

$$\boxed{3x + 4y + \sqrt{24}z = 1}$$